THREE PARADIGMS OF PHYSICAL WORLD MODELING

Contrasting Generative Pixel Diffusion, Latent State Transitions (JEPA/RSSM), and Deterministic SE(3) Coordinate Trees

1. GENERATIVE PIXEL DIFFUSION

Video Diffusion (Sora / Genie)

x_t (Pixels) + $a_t \rightarrow$ 50 Denoising Steps

Output: $\hat{x}_{t+1}$ (Rendered Video)

Computes all background pixels & lighting

Latency (P50/P99): **450 / 1200 ms** ❌

Memory Footprint: **16+ GB VRAM**

Compute Load: **> 30 TFLOPs**

FATAL EMBEDDED DEFECTS:

- **Pixel Hallucinations:** Objects phase through solid walls or change mass.
- **Freshness Wall Violation:** Latency exceeds 60 ms by 20x, forcing collisions.
- **Compute Starvation:** Consumes 100% SoC power, causing thermal throttle.

VERDICT FOR PHYSICAL AI:

✗ Inadmissible for Real-Time Control

2. LATENT DYNAMICS (JEPA / RSSM)

V-JEPA / DreamerV3 RSSM

$E_\theta(x_t) = z_t \in Z$ (Feature Space)

Transition: $\hat{z}_{t+1} = f_\varphi(z_t, a_t)$

Predicts semantics without pixel rendering

Latency (P50/P99): **2.2 / 3.8 ms** ✓

Memory Footprint: **420 MB DRAM**

Compute Load: **0.08 TFLOPs**

SYSTEMS ADVANTAGES:

- **150x Faster:** Skips high-overhead pixel decoders; executes directly on NPU.
- **Noise Invariance:** Ignores dynamic background clutter, shadows, and flicker.
- **Action Evaluation:** Rolls out 100 candidate trajectories in parallel < 5 ms.

VERDICT FOR PHYSICAL AI:

✓ Ideal for System 1.5 MPC Planning

3. METRIC SE(3) TREE + LIE ALGEBRA

Dynamic Transform Tree (SE(3))

$T_A^B = [\, R \mid p \,] \in SE(3)$, Covariance Σ

Tangent: $T(t) = T(t0) \cdot \exp(\hat{\xi}\, \Delta t)$

Zero singularities; exact metric coordinates

Latency (P50/P99): **< 0.05 ms (50 μs)** ★

Memory Footprint: **< 64 KB SRAM**

Determinism: **Zero Dynamic Allocations**

SYSTEMS ADVANTAGES:

- **100% Deterministic:** Runs inside 1 kHz FreeRTOS reflex loop on MCU.
- **Mathematical Safety Barrier:** Computes exact metric distance d_clear for CBF.
- **Sensory Lag Compensation:** Time-indexed buffer eliminates observation skew.

VERDICT FOR PHYSICAL AI:

✓ Mandatory for System 1 Safety Reflex